

Detailed derivation of STARWALL equations and matrices and extension to halo currents

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1 STARWALL equations without halo currents

1.1 Introduction

STARWALL is a code that provides the tangential field (B_{tan}) on the JOEREK boundary¹ given the normal field component ($\mathbf{B} \cdot \mathbf{n}$). In order to do that, the fields are expressed in terms of the currents using a Green's function method for the vector potential

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}'$$

where $\mathbf{J}(\mathbf{r}')$ is the current density flowing at the $\mathbf{r}'$ position.

In principle, the latter integral should be performed over all the plasma volume (apart from walls and coils). However, **the plasma volume integral is replaced by a surface integral on the plasma boundary, in which a virtual plasma current $\mathbf{J}_p$ flows**². As it will be shown, the virtual current $\mathbf{J}_p$ doesn't need to be calculated because it will disappear naturally from the equations.

The wall and other structures are represented also by surfaces using the thin wall approximation (see Appendix A). Therefore the problem is defined in terms of conducting shells that are separated by vacuum regions and that interact through the magnetic field.

1.2 General equations

First we find the electric field from Faraday's law in function of the electromagnetic potentials

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B} \quad \rightarrow \quad \mathbf{E} = -\partial_t \mathbf{A} - \nabla \Phi \quad (1)$$

Then we can use Ampère's law in the **Lorenz gauge** to find an expression of the vector potential in terms of the currents

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J} \quad \rightarrow \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' \quad (2)$$

Plugging this expression for the vector potential into (1) we obtain the equation of evolution for the currents

$$-(\mathbf{E} + \nabla \Phi) = \frac{\mu_0}{4\pi} \int_V \frac{\partial_t \mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' \quad (3)$$

which is our equation of motion requiring the next boundary conditions

At plasma-vacuum boundary S_p :	$\mathbf{E} = -\partial_t \mathbf{A}_{ctr} - \nabla \Phi$	(4)
At wall surface S_w :	$\mathbf{E} = \eta \mathbf{J} = \frac{1}{\sigma_w d_w} \mathbf{j}_w$, $\nabla \Phi = \nabla \Phi_w$	(5)
At coils surfaces S_{c_n} :	$\mathbf{E} = \eta \mathbf{J} = \frac{1}{\sigma_{c_n} d_{c_n}} \mathbf{j}_{c_n}$, $\nabla \Phi = \nabla U_{c_n}$	(6)

In the following, the subscripts p , w and c_n will respectively indicate that the quantity is evaluated at the plasma surface, wall surface and the n^{th} coil surface. Also σ stands for the electrical conductivity, d for the thickness of the thin structure and U_{c_n} is an external voltage given on the coils. In expressions (5) and (6) **the thin wall approximation has been used** (see Appendix A) which supposes that the current is constant in the direction perpendicular to the wall, which has a small width. Thus allowing us to write $\mathbf{J} = \mathbf{j}/d$ where $[\mathbf{j}] = \text{A/m}$ has dimensions of a linear density current.

¹We will refer to it also as "control surface"

²This is the virtual-casing principle, Shafranov 1972

Using the thin wall approximation, the volume integral in (3) can be written as

$$\int_V \frac{\partial_t \mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 \mathbf{r}' = \sum_{i=1}^{N_{surf}} \int \frac{\partial_t \mathbf{j}_i(\mathbf{r}')}{d_i |\mathbf{r} - \mathbf{r}'|} dS'_i dl'_i \approx \sum_{i=1}^{N_{surf}} \int \frac{\partial_t \mathbf{j}_i(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'_i \quad (7)$$

where the volume integral is decomposed as a sum of surface integrals after the integration in the direction normal to the i_{th} surface (dl_i). When using expression (7), the equation of motion (3) is

$$-(\mathbf{E} + \nabla \Phi) = \frac{\mu_0}{4\pi} \sum_{i=1}^{N_{surf}} \int \frac{\partial_t \mathbf{j}_i(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'_i \quad (8)$$

1.3 Boundary conditions

The boundary condition at the JOEREK-STARWALL interface is the continuity of the normal component of the magnetic field (due to $\nabla \cdot \mathbf{B} = 0$)

$$\mathbf{n} \cdot \mathbf{B}|_{JOEREK} = \mathbf{n} \cdot \mathbf{B}|_{ctr} \quad (9)$$

where $\mathbf{n} \cdot \mathbf{B}$ can be expressed in terms of the poloidal flux using the poloidal and toroidal coordinates (u, v) that describe the plasma-vacuum interface (control boundary)

$$\mathbf{n} \cdot \mathbf{B} = \mathbf{n} \cdot (\nabla \psi \times \nabla \varphi) = \frac{\mathbf{r}_u \times \mathbf{r}_v}{|\mathbf{r}_u \times \mathbf{r}_v|} \cdot (\nabla \psi \times \nabla \varphi) = \frac{2\pi}{|\mathbf{r}_u \times \mathbf{r}_v|} \frac{\partial}{\partial u} \psi(u, v) \quad (10)$$

$$(R(u, v), \varphi, Z(u, v)), \quad v = \varphi/(2\pi), \quad 0 \leq u, v \leq 1, \quad \mathbf{r}_{u/v} = \partial_{u/v} \mathbf{r}$$

where we have assumed that **the control boundary is axisymmetric** so $\mathbf{n} \cdot \nabla \varphi = 0$.

Thus the B.C. imposes the continuity of $\psi(u, v)$ and it's derivative along u at the interface. Then for reduced MHD, we can write the vector potential at the control surface as

$$\dot{\mathbf{A}}_{ctr} = 2\pi \dot{\psi} \nabla v \quad (11)$$

where ψ is now the JOEREK poloidal flux variable and the dot denotes the time derivative.

1.4 The finite element method

For the finite element method, **the currents are expressed in terms of current potentials ϕ in order to satisfy $\nabla \cdot \mathbf{j} = 0$** . The choice $\mathbf{j} = \mathbf{n} \times \nabla \phi(u, v)$ automatically satisfies the zero divergence³, where u and v are a poloidal and toroidal coordinates of a surface. For poloidally and toroidally closed tori, net current potentials must be added to single-valued periodic $\phi_s(u, v)$ potentials

$$\phi(u, v) = I^T u + I^P v + \phi_s(u, v) \quad (12)$$

where I^T and I^P are the net-toroidal and net-poloidal currents.

In the chosen finite element method, **the plasma, wall and other surfaces are discretized with triangles**. The position vector on a triangle is given by a local set of coordinates (α, β)

$$\mathbf{r} = \mathbf{r}_1 + \alpha \mathbf{r}_{2,1} + \beta \mathbf{r}_{3,1} \quad \alpha \geq 0, \quad \beta \geq 0, \quad 0 \leq \alpha + \beta \leq 1, \quad \mathbf{r}_{i,k} = \mathbf{r}_i - \mathbf{r}_k \quad (13)$$

where $i, k = 1, 2, 3$ number the vertexes of the triangles anti-clockwise.

The current potentials are chosen to vary linearly with respect the local coordinates, therefore the current in each triangle is constant and we can find the following expressions

³ $\nabla \cdot (\mathbf{n} \times \nabla \phi) = \nabla \phi \cdot \nabla \times \mathbf{n} - \mathbf{n} \cdot \nabla \times \nabla \phi = 0$ because $\mathbf{n} = \nabla S$ where $S = 0$ is the equation of the wall surface

$$\mathbf{j}_\Delta = \frac{\phi_1 \mathbf{r}_{2,3} + \phi_2 \mathbf{r}_{3,1} + \phi_3 \mathbf{r}_{1,2}}{|\mathbf{r}_{2,1} \times \mathbf{r}_{3,2}|} \quad \text{or} \quad \mathbf{j}_\Delta = \sum_{i=1}^3 \phi_i \mathbf{e}_i, \quad \mathbf{e}_i = \frac{1}{2} \epsilon_{ikl} \frac{\mathbf{r}_{k,l}}{|\mathbf{r}_{2,1} \times \mathbf{r}_{3,2}|} \quad (14)$$

where ϕ_i is the potential at vertex i of the triangle. With this discretization the surface integrals can be decomposed as sums of triangles integrals

$$\int \frac{\partial_t \mathbf{j}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS' = \sum_{m=1}^{N_{trian}} \int_{\Delta_m} \frac{\partial_t \mathbf{j}_{\Delta_m}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS' = \sum_{m=1}^{N_{trian}} \partial_t \mathbf{j}_{\Delta_m} \int_{\Delta_m} \frac{1}{|\mathbf{r} - \mathbf{r}'|} dS' \quad (15)$$

1.5 Plasma current potentials equations

In the plasma boundary, $-(\mathbf{E} + \nabla \Phi) = \dot{\mathbf{A}}_{ctr}$ and equation (8) becomes⁴

$$\frac{4\pi}{\mu_0} \dot{\mathbf{A}}_{ctr} = \int_{S_p} \frac{\dot{\mathbf{j}}_p(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'_p + \int_{S_w} \frac{\dot{\mathbf{j}}_w(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'_w \quad (16)$$

that takes the following discretized form by using (14) and (15)

$$\frac{1}{\mu_0} \dot{\mathbf{A}}_{ctr} = \frac{1}{4\pi} \sum_{i=1}^3 \left(\sum_{m=1}^{N_{triP}} \dot{\phi}_{idx(m,i)}^p \mathbf{e}_{m,i}^p \int_{\Delta_m} \frac{dS'_p}{|\mathbf{r} - \mathbf{r}'|} + \sum_{l=1}^{N_{triW}} \dot{\phi}_{idx(l,i)}^w \mathbf{e}_{l,i}^w \int_{\Delta_l} \frac{dS'_w}{|\mathbf{r} - \mathbf{r}'|} \right) \quad (17)$$

where $idx(m,i)$ is the index of the potential on triangle m at vertex i . (17) is then projected in order to obtain $N_{pot,p}$ scalar equations for the $N_{pot,p}$ different node plasma potentials. The projection is done with the operation $\sum_j \sum_{n=1}^{N_{triP}} \delta_{k,idx(n,j)} \int_{\Delta_n} dS_p \mathbf{e}_{n,j}^p \cdot$ and the equation looks like

$$\sum_j \sum_{n=1}^{N_{triP}} \delta_{k,idx(n,j)} \int_{\Delta_n} \frac{1}{\mu_0} \mathbf{e}_{n,j}^p \cdot \dot{\mathbf{A}}_{ctr} dS_p = \sum_{q=1}^{N_{pot,p}} M_{k,q}^{pp} \dot{\phi}_q^p + \sum_{r=1}^{N_{pot,w}} M_{k,r}^{pw} \dot{\phi}_r^w \quad (18)$$

where we have defined the matrices

$$M_{k,q}^{pp} \equiv \frac{1}{4\pi} \sum_{i,j=1}^3 \sum_{m,n=1}^{N_{triP}} \delta_{k,idx(n,j)} \delta_{q,idx(m,i)} \mathbf{e}_{m,i}^p \cdot \mathbf{e}_{n,j}^p \int_{\Delta_m} \int_{\Delta_n} \frac{dS'_p dS_p}{|\mathbf{r} - \mathbf{r}'|} \quad (19)$$

$$M_{k,r}^{pw} \equiv \frac{1}{4\pi} \sum_{i,j=1}^3 \sum_{n=1}^{N_{triP}} \sum_{m=1}^{N_{triW}} \delta_{k,idx(n,j)} \delta_{r,idx(m,i)} \mathbf{e}_{m,i}^w \cdot \mathbf{e}_{n,j}^p \int_{\Delta_m} \int_{\Delta_n} \frac{dS'_w dS_p}{|\mathbf{r} - \mathbf{r}'|} \quad (20)$$

1.5.1 Coupling term (LHS term) for reduced MHD

Equation (18) can also be obtained with the time integrated form of (16) (Ampère's law) and doing the operation $\frac{\partial}{\partial \phi^p} \int_{S_p} dS_p \mathbf{j}_p \cdot$ to the equation. This procedure allow us to compute the LHS of (18) with more accuracy by doing

$$\begin{aligned} \int_{S_p} \frac{\mathbf{A}_{ctr} \cdot \mathbf{j}_p dS_p}{2\pi} &= \int_{S_p} \psi \mathbf{j}_p \cdot \nabla v dS_p = \int_{S_p} \psi \nabla v \times \mathbf{n} \cdot \nabla \phi^p dS_p = \int \int \psi \frac{\partial \phi^p}{\partial u} du dv = \\ &= \boxed{I_T^p \int \int \psi(u, v) du dv - \int \int \phi_s^p(u, v) \frac{\partial \psi}{\partial u} du dv \approx \sum_p \sum_e M_{pe} \phi^p \psi^e} \end{aligned} \quad (21)$$

where we have used integration by parts for the singled-valued periodic potentials ϕ_s . When discretizing ϕ^p and ψ at the boundary, the matrix M_{pe} will give the contribution of the discretized component of the flux e to the plasma potential p . The LHS of (18) is finally found by doing $\partial_t \partial_{\phi^p}$ to (21)

⁴ With the formalism taken up to know, **we can include the coils and other elements as if they were part of the wall**. Therefore we will just distinguish between plasma and wall surface.

1.5.2 Final form for reduced MHD

Dropping the matrices indexes (without any change) and writing the potentials in a vectorial form the plasma current potentials equation becomes

$$M_{pp}\dot{\phi}^p + M_{pw}\dot{\phi}^w = M_{pe}\dot{\psi}\frac{2\pi}{\mu_0} \quad (22)$$

The linearity of the equation allows to express the plasma potentials in terms of the other variables.

$$\dot{\phi}^p = -M_{pp}^{-1}M_{pw}\dot{\phi}^w + M_{pp}^{-1}M_{pe}\dot{\psi}\frac{2\pi}{\mu_0} \quad (23)$$

1.6 Wall current potentials equations

The procedure to obtain the wall potentials equation is very similar to the plasma potentials. On the wall, equation (8) takes the form

$$-\left(\frac{1}{\sigma d}\mathbf{j} + \nabla\Phi\right) = \frac{\mu_0}{4\pi} \sum_{i=1}^{N_{surf}} \int \frac{\partial_t \mathbf{j}_i(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'_i \quad (24)$$

Discretizing in triangles and doing the operation $\sum_j^3 \sum_{n=1}^{NtriW} \delta_{k,idx(n,j)} \int_{\Delta_n} dS_w \mathbf{e}_{n,j}^w$, (24) becomes

$$M_{wp}\dot{\phi}^p + M_{ww}\dot{\phi}^w = -\frac{1}{\mu_0} R_{ww} N_{ww} \dot{\phi}^w - \frac{1}{\mu_0} V_{ww} \mathbf{U}^w \quad (25)$$

where we have defined the matrices

$$M_{wp} \equiv M_{k,r}^{wp} \equiv \frac{1}{4\pi} \sum_{i,j=1}^3 \sum_{n=1}^{NtriW} \sum_{m=1}^{NtriP} \delta_{k,idx(n,j)} \delta_{r,idx(m,i)} \mathbf{e}_{m,i}^p \cdot \mathbf{e}_{n,j}^w \int_{\Delta_m} \int_{\Delta_n} \frac{dS'_p dS_w}{|\mathbf{r} - \mathbf{r}'|} \quad (26)$$

$$M_{ww} \equiv M_{k,q}^{ww} \equiv \frac{1}{4\pi} \sum_{i,j=1}^3 \sum_{m,n=1}^{NtriW} \delta_{k,idx(n,j)} \delta_{q,idx(m,i)} \mathbf{e}_{m,i}^w \cdot \mathbf{e}_{n,j}^w \int_{\Delta_m} \int_{\Delta_n} \frac{dS'_w dS_w}{|\mathbf{r} - \mathbf{r}'|} \quad (27)$$

$$N_{ww} \equiv N_{k,q}^{ww} \equiv \sum_{i,j=1}^3 \sum_{n=1}^{NtriW} \delta_{k,idx(n,j)} \delta_{q,idx(n,i)} \mathbf{e}_{n,i}^w \cdot \mathbf{e}_{n,j}^w \int_{\Delta_n} dS_w \quad (28)$$

$$V_{ww} \equiv V_{k,q}^{ww} \equiv \sum_{i,j=1}^3 \sum_{n=1}^{NtriW} \delta_{k,idx(n,j)} \delta_{q,idx(n,i)} \mathbf{e}_{n,i}^w \cdot \mathbf{e}_{n,j}^w \int_{\Delta_n} dS_w \quad (29)$$

$$R_{ww} \equiv R_{i,j}^{ww} = \frac{1}{\sigma_i d_i} \delta_{ij} \quad (30)$$

and we have discretized the electric potential Φ to vary linearly on the triangles, so inside the triangle $\nabla\Phi \approx \sum_{j=1}^3 (U)^{idx(n,j)} \mathbf{e}_{n,j}^w$. Also R_{ww} is a diagonal matrix that takes into account the different resistivities that walls and coils can have.

1.6.1 Some words about the electric potential

If the electric potential is not given as a source (like an external voltage for the coils) an additional equation should be given in principle for the potential. However, if the wall separated by a vacuum, the wall potential is

$$\Phi^w(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV \quad (31)$$

which is very small compared to $\partial_t \mathbf{A}$ due to the considered time-length scales and can be neglected.

1.6.2 Similarity transform and final form

When replacing expression (23) into (25) we have

$$\mu_0 \tilde{M}_{ww} \dot{\phi}^w = -R_{ww} N_{ww} \phi^w - V_{ww} U^w - \tilde{M}_{we} \dot{\psi} 2\pi \quad (32)$$

where we have defined the matrices

$$\tilde{M}_{ww} \equiv M_{ww} - M_{wp} M_{pp}^{-1} M_{pw} \quad (33)$$

$$\tilde{M}_{we} \equiv M_{wp} M_{pp}^{-1} M_{pe} \quad (34)$$

For solving equation (32) we can decouple the variables with the similarity transform S

$$\phi^w \equiv \begin{pmatrix} \phi^{coils} \\ \phi^{wall} \end{pmatrix} = S \mathbf{Y} \quad (35)$$

such that⁵

$$S^T \tilde{M}_{ww} S = D \quad \text{and} \quad S^T R_{ww} N_{ww} S = I \quad (36)$$

where D is a diagonal matrix and I is the identity matrix. Therefore equation (32) becomes

$$\mu_0 \dot{\mathbf{Y}} = -M_{yy} \mathbf{Y} - M_{ye} \dot{\psi} - M_{yu} U^w \quad (37)$$

where we have defined the matrices

$$M_{yy} \equiv D^{-1} \quad (38)$$

$$M_{ye} \equiv 2\pi D^{-1} S^T \tilde{M}_{we} \quad (39)$$

$$M_{yu} \equiv D^{-1} S^T \tilde{V}_{ww} \quad (40)$$

writing the equation in JOREK units

$$\dot{\mathbf{Y}} = -M_{yy} \mathbf{Y} - M_{ye} \dot{\psi} - M_{yu} U^w \quad (41)$$

where the performed normalization is

$$\begin{aligned} t_{SI} &= t \sqrt{\mu_0 \rho_0} \\ \mathbf{Y}_{SI} &= \mathbf{Y} / \mu_0 \\ \sigma_{SI} &= \sigma \sqrt{\rho_0 / \mu_0} \\ \mathbf{U}_{SI} &= \mathbf{U} / \sqrt{\rho_0 \mu_0} \end{aligned}$$

1.7 Btan at the plasma interface

The magnetic field at the interface can be computed doing

$$\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A} = -\frac{\mu_0}{4\pi} \sum_{i=1}^{N_{surf}} \sum_{m=1}^{N_{trian_i}} \mathbf{j}_{i,\Delta_m} \times \nabla \int_{\Delta_m} \frac{1}{|\mathbf{r} - \mathbf{r}'|} dS'_i \quad (42)$$

The tangential component is found simply by projecting the field with the tangential vector to the interface $B_{tan} = \mathbf{B}(\mathbf{r}) \cdot \mathbf{v}_{tan}(\mathbf{r})$. Expressed in terms of the discrete current potentials

$$B_{tan}(\mathbf{r}_i) = \mu_0 M_{\parallel w} \phi^w + \mu_0 M_{\parallel p} \phi^p \quad (43)$$

⁵The matrix S can be found by solving the general eigenvalue problem $\tilde{M}_{ww} \mathbf{x} = \lambda R_{ww} N_{ww} \mathbf{x}$. S is constructed then by writing the eigenvectors in columns.

where we have defined the matrices

$$M_{\parallel w} \equiv M_{i,k}^{\parallel w} \equiv -\frac{1}{4\pi} \sum_{j=1}^3 \sum_{m=1}^{NtriW} \delta_{k,idx(m,j)} \mathbf{v}_{tan}(\mathbf{r}_i) \cdot \mathbf{e}_{m,j}^w \times \nabla \int_{\Delta_m} \frac{1}{|\mathbf{r}_i - \mathbf{r}'|} dS'_w \quad (44)$$

$$M_{\parallel p} \equiv M_{i,k}^{\parallel p} \equiv -\frac{1}{4\pi} \sum_{j=1}^3 \sum_{m=1}^{NtriP} \delta_{k,idx(m,j)} \mathbf{v}_{tan}(\mathbf{r}_i) \cdot \mathbf{e}_{m,j}^p \times \nabla \int_{\Delta_m} \frac{1}{|\mathbf{r}_i - \mathbf{r}'|} dS'_p \quad (45)$$

When replacing the plasma current potentials with expression (23) equation (43) becomes

$$B_{tan}(\mathbf{r}_i) = \mu_0 \tilde{M}_{\parallel w} \boldsymbol{\phi}^w + \tilde{M}_{\parallel e} \boldsymbol{\psi} \quad (46)$$

where

$$\tilde{M}_{\parallel w} \equiv M_{\parallel w} - M_{\parallel p} M_{pp}^{-1} M_{pw} \quad (47)$$

$$\tilde{M}_{\parallel e} \equiv 2\pi M_{\parallel p} M_{pp}^{-1} M_{pe} \quad (48)$$

when normalizing into JOREK units and using the similarity transform we finally have

$$B_{tan}(\mathbf{r}_i) = M_{ey} \mathbf{Y} + \tilde{M}_{\parallel e} \boldsymbol{\psi} \quad (49)$$

where

$$M_{ey} \equiv \tilde{M}_{\parallel w} S \quad (50)$$

As a last step the function $B_{tan}(\mathbf{r}_i)$ should be transformed from the real space to the bezier space used in JOREK.

2 Extension for halo currents

2.1 Electric potential equation for a 3D wall

If the wall enters in direct contact with the plasma, $\nabla\Phi \cdot \mathbf{n}$ may be significant at the plasma-wall interface and the potential should be considered. By taking the divergence of Ohm's law we find

$$\nabla \cdot \mathbf{J} = 0 = \nabla \cdot (\sigma \mathbf{E}) = -\nabla \cdot (\sigma \nabla \Phi) - \partial_t \mathbf{A} \cdot \nabla \sigma \quad (51)$$

Therefore for a constant wall resistivity expression, (51) becomes the Laplace equation $\Delta\Phi = 0$ that can be solved with the Neumann boundary conditions $\nabla\Phi \cdot \mathbf{n}$ given by the continuity of the normal current

$$\mathbf{n} \cdot \mathbf{J}|_{plasma} = \mathbf{n} \cdot \mathbf{J}|_{wall} \quad (52)$$

2.2 Electric potential equation for a thin wall

In the thin wall approximation, the quantities are constant in the direction perpendicular to the wall, so $\mathbf{j}(u, v)$ and $\Phi^w(u, v)$. Moreover, the currents are always tangential to the wall and the halo currents can be considered as a tangential current source

$$\nabla \cdot \mathbf{j}(u, v) = S_{halo} \quad (53)$$

where $S_{halo} = \mathbf{n} \cdot \mathbf{J}|_{plasma}$ as it will be justified in the next section.

Then the form of equation (51) is

$$\boxed{\nabla \cdot (\sigma d \nabla \Phi(u, v)) + \partial_t \mathbf{A} \cdot \nabla (\sigma d) = -\mathbf{n} \cdot \mathbf{J}|_{plasma}} \quad (54)$$

2.2.1 Halo current source in terms of the normal current

S_{halo} will be given directly from the condition $\nabla \cdot \mathbf{J} = 0$ for a thin 3D wall. In a simplified picture, we can represent the wall as a slab in Cartesian coordinates with $\mathbf{e}_z$ being the normal direction to the wall surface. Then the 3D current can be written as

$$\mathbf{J}(x, y, z) = \frac{\mathbf{j}(x, y)}{d_w} + \mathbf{n} \cdot \mathbf{J}|_{plasma} \left(1 - \frac{z}{d_w}\right) \mathbf{e}_z \quad (55)$$

where we have chosen a linear dependence for the current in the normal direction in a way that is zero at the outer wall surface and $\mathbf{n} \cdot \mathbf{J}|_{plasma}$ at the plasma interface. The condition of zero divergence with this choice gives

$$\nabla \cdot \mathbf{j}(x, y) = \mathbf{n} \cdot \mathbf{J}|_{plasma} \quad (56)$$

therefore

$$S_{halo} = \mathbf{n} \cdot \mathbf{J}|_{plasma} \quad (57)$$

2.3 Decomposition of currents

If the term $\partial_t \mathbf{A} \cdot \nabla (\sigma d)$ is neglected from equation (54), we can add a non-zero divergence term to the current decomposition

$$\boxed{\mathbf{j}_w = \mathbf{n} \times \nabla \phi - \sigma d \nabla \Phi} \quad (58)$$

which will automatically satisfy (53) if the equation $\nabla \cdot (\sigma d \nabla \Phi(u, v)) = -\mathbf{n} \cdot \mathbf{J}|_{plasma}$ has been previously solved. With this form equation (24) will be

$$-\frac{1}{\sigma d} \mathbf{n} \times \nabla \phi = \frac{\mu_0}{4\pi} \int \frac{\partial_t (\mathbf{n} \times \nabla \phi - \sigma d \nabla \Phi)(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS'_w + \frac{\mu_0}{4\pi} \int \frac{\partial_t \mathbf{j}_p(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS_p \quad (59)$$

and we can see that in the RHS the halo currents will have an influence on the free-divergent currents.

2.4 Open questions

Should we also decompose the plasma currents $\mathbf{j}_p$ also with a term $\sigma d\nabla\Phi$ to conserve the current ?

3 Appendix A: Validity of thin wall approximation (Copied from Hoelzl 2010)

We will discuss the validity of the thin wall approximation in the following, showing that it is a reasonable model in most situations. For the discussion, we will consider a typical time scale τ of the perturbation, which can either be the inverse of its rotation frequency or the inverse of an instability growth rate. A magnetic field varying on the characteristic time scale τ penetrates into a conductor limited to the skin depth $\delta_{skin} = \sqrt{2\eta\tau/(2\pi\mu_r\mu_0)}$, where μ_r is the relative permeability. ITER wall structures are mainly built from austenitic steel, and even martensitic components will be deep in the saturated regime to the large toroidal field, such that $\mu_r = 1$ can be taken in estimates. For the steel walls ITER will have ($\eta \approx 10^{-6}\Omega m$) we obtain $\delta_{skin} \approx 0.5m\sqrt{\tau[s]}$. The characteristic time for a magnetic perturbation to become homogeneous across the $d = 6cm$ thick ITER walls is given by $\tau_{skin} = 7ms$. An external kink mode which can fully be stabilized by an ideally conducting wall, but is unstable in the presence of a resistive wall is called a resistive wall mode. It's growth time τ is approximately given by the characteristic time L/R time τ_{wall} of the wall structures, which for the ITER conducting vessel is of the order of 200 ms. For such a mode with $\tau \approx \tau_{wall} \gg \tau_{skin}$, the field is homogeneous across the thin wall and the approximation is well justified.

The opposite limit of an instability that is weakly coupled to the walls and can not be stabilized by ideally conducting walls corresponds to $\tau \ll \tau_{skin} \ll \tau_{wall}$. In this case, the magnetic perturbation can only penetrate into a very thin surface layer of the conductors. In principle, an effective - increased - resistivity needs to be considered for the wall current evolution. However the wall completely shields the magnetic field associated to such a fast mode and effectively acts as an ideally conducting wall - however with little influence on the mode growth. So the correction of the resistivity is not necessary and the approximation is well justified.

For the remaining case $\tau \approx \tau_{skin}$ where the magnetic perturbation has partially penetrated through the wall. This situation corresponds to an increased wall resistivity, which now has to be taken into account but the non-linear behaviour might not be well calculated. Tokamak instabilities with such a growth rate are, however, rather untypical since the wall coupling must be very marginal.

From the geometrical point of view, the thin wall approximation only plays a role if the mode wall distance is smaller than the wall thickness.